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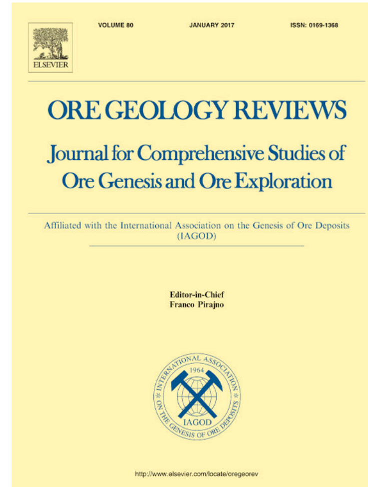

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In situ compositional and sulfur isotopic analyses of banded sphalerite from the Kangjiawan Pb-Zn deposit, South China: Insights into incorporation mechanisms of critical metals (Cd, Ga, In)

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ABSTRACT

The critical metals Cadmium (Cd), gallium (Ga), germanium (Ge) and indium (In) are essential to the development of the green technologies that are at the core of low-carbon societies. Although sphalerite is their principal host mineral in nature, high-temperature hydrothermal Pb-Zn deposits have seldom been the subject of focused studies on these metals. Using in situ measurements of major elements by EMPA and trace elements and sulfur isotopes by LA-ICP-MS, together with LA-ICP-MS elemental mapping, this study investigated the microscale critical elemental distributions and incorporation mechanisms in banded sphalerite from the Kangjiawan skarn Pb-Zn deposit, South China. Three typical optical zoning bands can be identified (bands R1 to R3 from dark to light). Different bands are characterized by different concentrations of minor/trace elements, with the darker bands displaying higher Fe, Mn and Cu concentrations. Spot concentrations for Cd, Ga and In range from 0.21 to 0.29 wt%, 1.03 to 41.40 ppm, and 0.12 to 10.14 ppm, respectively. Cd is homogeneously distributed within the sphalerite grains, whereas Ga is mainly concentrated in band 2. Rhythmic distribution of In concentrations notwithstanding, it does not correlate with the above colour banding. Moreover, elemental concentrations do not display any systematic increase/decrease from core to rim at the micrometer-scale, which indicates that microscale processes at or near the growing crystal edges account for the observed compositional zoning. The Kangjiawan sphalerite displays $\delta^{34}S_{V\text{-}CDT}$ values ranging from 1.98 to 5.18 ‰, typical of magmatic-derived ore-forming materials. Sulfur isotopes also do not display systematic variations from core to rim, further highlighting that the banded sphalerite should form by intrinsic, self-organized processes. The elemental correlation analyses suggest the isovalent substitution $Cd^{2+} \leftrightarrow Zn^{2+}$, and coupled substitution $In^{3+} + (Cu, Ag)^{+} \leftrightarrow 2Zn^{2+}$. A strong correlation between Ga and Mn is manifested, probably suggesting a more complex mechanism $Ga^{3+} + Mn^{2+} + (Cu, Ag)^{+} \leftrightarrow 3Zn^{2+}$. Combined with published global sphalerite data, this study advocates the more promising of MVT deposits for Ga and Ge exploration whereas skarn deposits for In exploration. There is no distinction between low- and high-temperature hydrothermal deposits in forming economic Cd mineralization. Strikingly, the Kangjiawan sphalerites are characterized by high Ga contents, similar to those from MVT deposits. Collectively, our study highlights the potential of skarn Pb-Zn deposits, at least from South China, for not only Cd and In exploration, but also Ga exploration.

Keywords: critical metals; banded sphalerite; skarn; Kangjiawan; South China

INTRODUCTION

Cadmium (Cd), gallium (Ga), germanium (Ge) and indium (In), which were regarded as the critical metals for the sustained development, are widely used in high-tech fields (e.g., solar energy industry, semiconductor industry and biomedical applications) (Le Goff et al., 2019). These metals have thus been listed in the critical raw materials by the major economic community, such as China, the European Union (EU) and the United States Geological Survey (USGS) (USGS, 2020; European Commission, 2020). Due to their importance, further understanding regarding its geologic cycle is necessary (Cooke et al., 2011; Belissont et al., 2014; 2015; 2016; Frenzel et al., 2016; Huston and Bastrakov, 2024). Sphalerite, one of the most important industrial minerals of Zn, can effectively accommodate critical metals including In, Cd, Ga and Ge (Belissont et al., 2014; Cook et al., 2009; Liu et al., 2022). Although a substantial amount of research has been carried out on sphalerite in the Pb-Zn polymetallic deposits, the endowment of the critical metals in sphalerite is rarely reported (Fougerouse et al., 2023). Therefore, a complete understanding of the distributions of these critical metals, as well as their incorporation mechanisms into the principal host minerals (i.e., sphalerite), is still lacking (Huston and Bastrakov, 2024).

Sphalerite is a common mineral in skarn deposits (Meinert et al., 2005; Shen et al., 2022), carbonate-replacement Mississippi Valley-type (MVT) deposits (Guan et al., 2024), vein-type deposits, porphyry-epithermal deposits (Swinkels et al., 2021) and massive sulfide deposits (Mishra et al., 2021). Previous studies in sphalerite showed that high Ge and Ga contents generally could be linked to the MVT deposits (Guan et al., 2024; Cook et al., 2009; Huston and Bastrakov, 2024; Fougerouse et al., 2023), which have been the focus of current high-tech critical metal studies. In contrast, micron-scale distributions of high-tech critical metals regarding sphalerite from porphyry and skarn deposits are less investigated. There is no doubt that the magmatism-related high-temperature hydrothermal deposits (e.g., porphyry deposits, skarn deposits) are important sources of global Zn (and thus sphalerite) (Ma et al., 2025). It is thus indispensable for a better understanding of the distribution regularity and involved geological setting of critical metals in the crust. Besides, many substitution mechanisms for trace and minor elements in sphalerite from low-temperature hydrothermal deposits have been proposed (Belissont et al., 2014), whereas whether they apply to other Zn deposits is unclear.

This impedes our knowledge of mechanisms controlling the incorporation and redistribution of critical elements in sphalerite. These considerations provided the major motivation for this work.

Integrated geochemical and mineralogical studies of sphalerite from the Kangjiawan large skarn Pb-Zn polymetallic deposit have been extensively reported (Shen et al., 2022; Shu et al., 2024; Zhang et al., 2024). Although none of Cd, Ga, Ge and In are extremely enriched in this deposit, previous studies have revealed that sphalerite from this deposit displays zoned elemental patterns and higher Cd and Ga contents compared to common skarn deposits (Shen et al., 2022; Shu et al., 2024). Thus, sphalerite from this deposit is a key target for investigations into the enrichments of Cd, Ga, and In, as well as related minor and trace elements in high-temperature hydrothermal deposits.

In this study, in situ elemental and sulfur isotopic compositions together with elemental mapping were measured by the EMPA and LA-ICP-MS methods. This work aims to resolve how critical elements (e.g., In, Cd, Ga, Ge) were distributed and incorporated into sphalerite under high-temperature hydrothermal environments as well as provide useful guidelines on the economic potential of high-tech critical metals in skarn deposits.

GEOLOGICAL SETTING

Regional geology

The Kangjiawan deposit is situated in the county of Changning, about 210 km south of the provincial capital of Hunan in South China. It is a typical deposit from the western part of the Shuikoushan large Pb-Zn-Au polymetallic ore field, which is located at the middle part of the NNE-trending Qinzhou-Hanzhou Cu-Fe-W-Mo-Pb-Zn Metallogenic Belt (QHMB; Fig.1), which is also a Neoproterozoic conjunction zone between the Yangtze Craton and Cathaysia Block (Chen and Jahn, 1998; Li et al., 2003; Wang et al., 2007; Mao et al., 2013). It extends along the southeast of the Youjiang Basin, a famous basin in South China that preserves the records of the tectonic history from Tethyan, through Paleo-Pacific to Tibetan domains (Wang et al., 2020). This belt is characterized by a Mesoproterozoic meta-igneous basement and was reactivated by the orogeny in the Early Paleozoic and subduction accretion of the Paleo-Pacific in the Mesozoic.

The QHMB, a key metallogenic belt, hosts numerous giant magmatism-related porphyry-skarn Cu-Mo-W-Sn deposits, as well as Pb-Zn-Au (Ag) polymetallic deposits, such as Shuikoushan, Kangjiawan, Shizhuyuan, Huangshaping-Baoshan, and Lengshuikeng deposits (Zhao et al., 2017; Li et al., 2019) (Fig. 1B). Previous studies have determined three mineralization series: (1) the Neoproterozoic spilitic keratophyre formation-related VHMS Cu deposits (e.g. Xiqiu, Luocheng, Tieshajie deposits), which are limitedly distributed in the Pingxiang area and formed at ~850Ma (Luo et al., 2006); (2) the mid-late Jurassic granite-related porphyry-skarn-veined W-Sn-Cu-Mo-Pb-Zn-Au-(Ag) deposits; and (3) the Cretaceous epithermal Sn-(W)-Au-Ag deposits (~140Ma), located at the west part of QHMB. Moreover, because in the middle QHMB host rocks for polymetallic orebodies are dominated by carbonate rocks, skarn deposits are the predominant deposit type in this area. This is different from the north and south QHMB, where the host rocks are dominated by metaclastic rocks. Therefore, in these areas quartz vein-type and volcanic hydrothermal deposits are more common than skarn deposits (Mao et al., 2013).

The Shuikoushan ore field, hosts approximately 3 Mt of Pb and Zn resources and 70 t of Au resources (Qin et al., 2022), is mainly composed of the large Kangjiawan deposit (Pb, Zn, Au, Ag), the large Laoyachao deposit (Fe, Cu, Pb, Zn, Au), the large Yagongtang deposit (Fe, Cu, Pb, Zn), and the Longwangshan deposit (Au), all of which are predominantly characterized by skarn mineralization (Huang et al., 2015). In these deposits, vein-type and brecciated Pb-Zn polymetallic mineralization is also locally observed. The skarn and veined Zn-Pb polymetallic orebodies are generally hosted in the carbonate rocks of the Permian Qixia Formation (P_1q) and, to a lesser extent, the shale units of the Dangchong Formation (P_1d). The structures developed in the ore field are mainly NE-oriented, NNE-oriented and EW-oriented. Due to the development of the early Mesozoic EW-direction orogenic compression between the Indochina block and South China block, a series of N-S-oriented folds and faults formed. It also allows the emplacement of multiple intrusions (granodiorite, dacite porphyry, granite porphyry) along deeply penetrating faults and covering an area of approximately 4.8 km^2. Among them, granodiorite is volumetrically the most abundant igneous rock in the Shuikoushan ore field, with an outcrop exposure accounting for ca. 1.8 km^2. Extensive skarn-, hydrothermal vein- and breccia-type ores were developed along the contact zones between the granodiorite and the Qixia and Dangchong Formations. Other intrusions (e.g., granite porphyry) are volumetrically much smaller than granodiorite and have been confirmed to be unrelated to polymetallic mineralization of the Shuikoushan ore field (Gong, 2012; Yang et al., 2016; Qin et al., 2020) (Fig. 2A).

Deposit geology

The Kangjiawan Pb-Zn deposit, with a proven ore reserve of 1.96 Mt at 4.70 % Zn and 6.02 % Pb (Shen et al., 2022), is the largest skarn Pb-Zn deposit in the Shuikoushan ore field (Chang et al., 2019). The strata exposed in the south of the ore field contain thick beds of limestone and dolomitic limestone of the Carboniferous Shidengzi Formation (C_2s) and the Hutian Group ($C_{2+3}ht$). The late Paleozoic strata include dark blue-gray bedded limestone and siliceous shale belonging to the Qixia Formation (P_1q) and the Dangchong Formation (P_1d), as well as the coal-bearing carbonaceous shale of the Dangchong Formation (P_1d). The Qixia Formation (P_1q) and the Dangchong Formation (P_1d) host a substantial amount of skarn-type and vein-type Au, Fe, Cu, Pb, and Zn ores (Fig. 2B). Additionally, the sandstone is also exposed in the northern ore field, which belongs to the Jurassic Gaojiatian Formation (J_1g) and the Cretaceous Dongjing Formation (K_1d) (Fig. 2C).

Fifty-six veined Pb-Zn-Au orebodies occur in the Kangjiawan deposit. Three types of hydrothermal alteration zones (i.e., the skarn zone, breccia and silicification zone, and carbonatation zone) have been identified. The skarn zone is distributed in an area of approximately 700 m below the surface, and is characterized by a small amount of disseminated andradite, magnetite, pyrite and chalcopyrite. The andradite commonly suffered from intense low-temperature alteration with the occurrence of chlorite and calcite. The breccia and silicification zone developed in the fracture zone of the Permian detrital rocks and carbonate rocks. The silicified breccia was cemented by quartz-pyrite veins and veined Pb-Zn orebodies. The carbonatation zone extensively occurred in the carbonate rocks of the Qixia Formation, with re-crystallized calcite veins cutting the pre-existing Pb-Zn orebodies.

The main orebodies consist of stratabound veins and lenses, which are mainly found within the breccia and silicification zone. Mineralization is spatially and temporally associated with NS-trending faults, and to a lesser extent, NE-, NW- and EW-trending faults. Apart from faults, folds are also common in the Kangjiawan deposit, with fold orientations mostly trending from north to northeast. Mineralization and alteration can be observed in the hinge zone and limbs of these folds. The surface outcrop of igneous rocks is not exposed in the Kangjiawan deposit. Besides, no direct relationship has been discovered between orebodies and intrusive rocks at depth. Recently, skarn mineralization has been newly reported in the Kangjiawan deposit (Qin et al., 2020; 2022; Wang et al., 2024). It is confirmed that the Kanjiawan deposit should belong to the distal skarn Pb-Zn-Au-(Ag) metallogenic system (Qin et al., 2022). The dominant economic minerals of the Kangjiawan deposit comprise sphalerite (Fig. 3). Apart from the sphalerite, pyrite, chalcopyrite, natural gold, galena, natural silver, silver tellurite, pyrrhotite, hematite, arsenopyrite, bornite, chalcogenite, silver tetrahedrite and arsenopyrite also occur in subordinate amounts.

Qin et al. (2020; 2022) have confirmed that there are four hydrothermal-mineralization stages in the Shuikoushan ore field. According to the field relationships, only three hydrothermal stages are observed in the Kangjiawan deposit (Fig. 4). The first stage (skarn stage) corresponds to the development of skarns by the contact metamorphism and/or metasomatism processes. Unlike those in the Laoyachao and Yagongtang deposits, the skarn in Kangjiawan is generally barren. It has a monotonous mineral assemblage that includes garnet (andradite) and magnetite, with only a very small amount of pyrite, and chalcopyrite scattered in the crystal fissures (Fig. 3A, E, G-H). The second stage (hydrothermal veined stage) formed a large number of quartz-pyrite and quartz-polymetallic sulfide veins (Fig. 3B-D, F, I-M). Quartz-pyrite veins filled the zone of silicified breccia in the Douling Formation, with the extensive development of numerous pyrite rather than sphalerite and galena (Fig. 3B, I). In contrast, in the Kangjiawan deposit, most Pb-Zn ores, as well as many Au- and Ag-bearing sulfides, formed during the quartz-polymetallic sulfide veins stage. The third stage (late hydrothermal stage) is superimposed on the skarn stage and the hydrothermal veined stage, and formed a lot of low-temperature minerals, such as calcite, quartz, and chlorite (Fig. 3C, E, G).

MATERIALS AND METHODS

The sphalerite-bearing ore sample (KJW-1) was collected from the No. 12 mine face of the Kangjiawan deposit (Fig. 2B), which is about 385 m deep. This sample was taken from an economic Pb-Zn orebody where the sphalerite formed during the second mineralization stage mentioned above. Finally, four typical sphalerite crystals (Sph-1, Sph-2, Sph-3 and Sph-4) with well-developed rhythmic color banding were selected for the analyses below (Fig. 5). Sph-1 was only used for element mapping whereas the other three grains have been selected for quantitative element/isotope analyses.

The EMPA was conducted for the major compositions of the sphalerite. Laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) analyses were conducted to collect both quantitative point analyses and the maps for selected minor/trace elements of the Kangjiawan sphalerite samples. To reveal the sources of the ore-forming materials in the deposit, in situ sulfur isotope analyses were also conducted for the Kangjiawan zoned sphalerite. Detailed descriptions of these analytical methods, as well as the analytical precisions, were provided in the Appendix Method Description.

RESULTS

Sphalerite zoning patterns

Optical microscope observations of the sphalerite on 30 μm-thick polished sections revealed well-developed crystals heterogeneously coloured. Most sphalerite grains are millimeter-sized twinned crystals and show polygonal shapes. In the studied sample from the Kanjiawan deposit, sphalerite displays noticeable rhythmic banding. Three typical optical colour zoning patterns can be identified. (i) The first one is the dark brown to black band (R1), with a bandwidth of 15 to 25 μm. This band becomes lighter in colour and thinner in width from the core to the rim. Small amounts of galena, pyrite and chalcopyrite micro-inclusions can be locally found within this band. (ii) The second one is the reddish-brown band (R2) that has a stable zonal field with a width of 20-30 μm, except for the crystal cores where the band width can be 200-500 μm in diameter. In the Kangjiawan deposits, all the observed sphalerite crystal cores are characterized by the band R2. Besides, the band R2 in the cores is generally irregular in shape, indicating the post-nucleation fluid metasomatism. (iii) The last band (R3) is light yellow to colourless and generally 20-40 μm in width. Different from bands R1 and R2, the width of the band R3 in some crystals becomes wider from the core to the rim, although the colour of this band also turns lighter (Fig. 3).

Minor/trace element distribution

Micrographs under transmitted light and LA-ICP-MS maps of typical elements for sphalerite Sph-1 are shown in Figure. 4. Detailed observation of the zoning patterns in the studied sphalerite grain shows that although the distributions of Fe concentrations tend to correlate with the colour of rhythmic bands, with the band R3 displaying higher Fe concentrations, it does not seem to be the main cause of rhythmic colour bands. In contrast, Figure 4 shows that Mn and Cu concentrations correlate well with the rhythmic bands, with the darker bands R1 and R2 characterized by higher Mn and Cu concentrations than the band R3. Similar to Mn and Cu, Ga and In also display rhythmic distributions. The differences include that Ga is mainly concentrated in the band R2, with R1 and R3 displaying no visible difference to the background values, and that the distribution of In does not correlate with the colour banding. In contrast, the distribution of Cd does not display visible correlations with the rhythmic bands. Zn is also homogeneously distributed within the sphalerite grains. Corresponding to previous studies (Shen et al., 2022), sphalerite from the economic mineralization stage of the Kangjiawan deposit is characterized by low Ge contents and is nearly undetectable within the LA-ICP-MS maps. The same case is also observed for Pb. The distributions of Ag and Sb are mainly concentrated in the micron-scale fractures of the studied sphalerite grains, indicating that they were precipitated from a later fluid. All these collectively highlight complex relations between the colours of rhythmic bands and element distributions.

Minor/trace element contents

Three grains (Sph-2, Sph-3 and Sph-4) were used for in situ chemical composition analyses.

The full EMPA analysis data were shown in Appendix Table S1. Three grains display similar features. Averaged Zn concentrations of band R1 for three grains range from 58.07 to 59.58 wt%, lower than R2 and R3 (54.92 to 61.05 wt% and 62.03 to 63.88 wt%, respectively). In contrast, S concentrations do not display noticeable distinctions, which range from 33.14 to 33.45 wt% for R1, 32.89 to 33.24 wt% for R2, and 32.88 to 33.16 wt% for R3. Sphalerite from the Kangjiawan deposit shows significantly high Fe concentrations. R1 is composed of 6.08 to 10.24 wt% Fe, which is higher than R2 (4.98 to 5.50 wt%) and R3 (2.13 to 3.90 wt%). Moreover, Cd is detected in all analyzed spots and does not show noticeable distinctions among the three bands, with contents ranging from 0.21 to 0.29 wt%. Out of 86 analyzed spots, Cu was detected in 51 spots and Mn was consistently detected in the analyzed spots. Both Cu and Mn concentrations are highest in R1 and lowest in R3. This characteristic is consistent with the rhythmic patterns of Mn and Cu by the LA-ICP-MS mapping. It is noted that despite rhythmic patterns, from core to rim, a systematic increase/decrease is observed for none of the measured elements (Fig. 7).

The minor/trace elements analysis result is listed in Appendix Table S2. Figure 8 shows histograms illustrating the concentrations of elements relative to zoning patterns. 51 spots from three samples has also been selected for the PCA analyses to investigate the correlativity between chemical component and the bandings of sphalerite (Fig. 9). Only analyses with smooth ablation profiles (implying a homogeneous distribution of the elements) were retained. Sphalerite from the Kangjiawan deposit displays very low Ge concentrations, which range from below the limit of detection to 2.25 ppm. In contrast, In and Ga display much higher concentrations from 0.12 to 10.14 ppm (averaging 2.80 ppm) and 1.03 to 41.40 ppm (averaging 12.87 ppm), respectively. In is mostly concentrated in band R3 compared to bands R1 and R2, whereas Ga is much more concentrated in band R2, which is also supported by the PCA diagrams. Moreover, the analyzed grains display significant Fe and Cd concentrations (1.59 to 5.66 wt% and 2449 to 4587 ppm, respectively). According to EPMA and element mapping results, Fe concentrations exhibit rhythmic variations across the three bands and are significantly enriched in band R1. Mn and Cu concentrations vary by two orders of magnitude (322 to 10236 ppm and 21 to 6375 ppm, respectively) and are particularly concentrated in band R1. Sn also shows a wide range of concentrations (6 to 2926 ppm) and is evidently concentrated in R1, similar to Mn and Cu. Similar to EMPA results, none of these trace and minor elements display a systematic increase/decrease from core to rims (see Table S2).

Sulfur isotopes

The in situ sulfur isotopic result for the banded sphalerite (Sph-2) is listed in Appendix Table S3 and shown as histograms in Fig. 10A-C. $\delta^{34}S_{V\text{-}CDT}$ values vary from 1.98 to 5.18 ‰, and slight variation could be observed by detailed analysis of each band. $\delta^{34}S_{V\text{-}CDT}$ values of band 1 range from 1.98 to 5.18 ‰, with a mean value of 3.73 ‰ (n = 4). $\delta^{34}S_{V\text{-}CDT}$ values of bands 1 and 2 range from 2.80 to 4.51 ‰ and 2.57 to 3.88 ‰, respectively, with band 1 (3.80 ‰) characterized by a higher mean value than band 2 (3.35 ‰). To further indicate the sulfur isotope variations during the growth of banded sphalerite, Figure 10D also shows the variations of the averaged $\delta^{34}S_{V\text{-}CDT}$ value of individual bands. There are no systematic variations of sulfur isotopes from core to rim for the same bands, with the sulfur isotopes for each band close to the weighted mean value (3.63 ± 0.64 ‰).

DISCUSSION

Genesis of zoned sphalerite from the Kangjiawan deposit

Elemental distribution and substitution mechanism

Previous studies show that sphalerites in many Pb-Zn deposits are characterized by two main typical optical colour zoning patterns (Belissont et al., 2016): rhythmic banding characterized by alternating parallel dark and light growth bands from the core to the rim, and sector zoning occurring as crosscutting dark zones. The sector zoning generally consists of a series of small and parallel truncation planes highly variable in size, and accounts for their highly indented boundaries (Belissont et al., 2014). In the Kanjiawan deposit, sphalerite displays noticeable rhythmic banding, although the sector zoning is not observed. As clearly indicated by the trace and minor elements mapping, the various color bands within the studied sphalerites are characterized by varying concentrations of minor and trace elements. For example, darker bands R2 and R3 display higher Fe, Mn and Cu concentrations than light band R3. Although Ga and In also display rhythmic distributions, Ga is mainly concentrated in the band R2, and the distribution of In seemingly does not correlate with the color banding. This colorful and thus compositional zoning has been recorded not only in sphalerite, but also in pyrite, arsenopyrite, andradite and zircon. Generally, such compositional zoning could be explained by certain self-organized non-equilibrium processes occurring during crystal growth. As summarized by Belissont et al. (2014), the formation of banding patterns could be explained by two processes, including (i) the superimposed effect of multiple fluids with distinct compositions in an open system, and (ii) self-organized non-equilibrium processes (microscale variations) in a relatively closed system. At the micrometer-scale, the major and trace elements of the Kangjiawan banded sphalerite do not display any regular variation (increase/decrease) from the core to the rim. This indicates that the compositional zoning would be attributed to microscale processes at or near the growing crystal edges. The consistent ore-forming material supplies during the growth of the Kangjiawan sphalerite, indicated by in situ sulfur isotope analyses (see section 5.1.2), further support that the banded patterns of the Kangjiawan sphalerite should be triggered by intrinsic, self-organized processes (Katsev and L' Heureux, 2001; Di Benedetto et al., 2005; Belissont et al., 2014).

In sphalerite from the Kangjiawan deposits, spot concentrations for Fe, Mn and Cd range from 0.33 to 12.36 wt%, 322 to 12419 ppm and 2449 to 4587 ppm, respectively, which indicates strong substitution of these elements into sphalerite. These results support the incorporation of Fe, Mn and Cd mainly via the isovalent substitution mechanisms ($Zn^{2+} \leftrightarrow Fe^{2+}$, $Zn^{2+} \leftrightarrow Mn^{2+}$, and $Zn^{2+} \leftrightarrow Cd^{2+}$), which has already been discussed extensively in several studies. Figure. 11A display a strong correlation between Ga + In + Ge + Sn +Sb +As and Cu + Ag. This is also observed in many studies, which has been generally highlighted as evidence of incorporation of tri- and tetravalent cations (e.g., In^{3+}, Sb^{3+}, Ga^{3+}, As^{3+}, Sn^{4+}, Ge^{4+}). The Cu and Ag are monovalent cations to enable the valence state to reach equilibrium in the coupled substitutions process. It is noted that both Sn and Ge are redox variable elements (Niu et al., 2025). Sn and (Ag, Cu) display a noticeable correlation, which has a trend sub-parallel to the 1:1 line (Fig. 11B). Neither Sn^{2+} nor Sn^{4+} can adequately explain this observed correlation, nor can they account for the high Sn concentrations found in the Kangjiawan sphalerite. Belissont et al. (2014) suggested that although the main oxidation states of Sn are +2 and +4, it could also be +3. If this is correct, the observed correlation in Figure 11B could be explained by the coupled substitution $Sn^{3+} + (Cu, Ag)^{+} \leftrightarrow 2Zn^{2+}$. Since Ge^{2+} could more easily enter into sphalerite (via the simple substitution of Zn^{2+}) than Ge^{4+}, Ge in the Kangjiawan sphalerite is more likely present in the Ge^{4+} state, considering the low Ge contents for the studied sphalerite. Belissont et al. (2016) recently confirmed that the Ge^{4+} is likely involved the following substitution: $Ge^{4+} + \square$ (vacancy) $\leftrightarrow 2Zn^{2+}$ or $Ge^{4+} + 2(Cu, Ag)^{+} \leftrightarrow 3Zn^{2+}$.

An interesting finding is that Ga displays an evidently positive correlation with Mn. Ga can only exist as Ga^{3+} (Fig. 11C). Although Mn could be Mn^{2+} and/or Mn^{4+} in many minerals, Bernardini et al. (2004) demonstrated that in sphalerite, Mn is present in the Mn^{2+} state. This suggests that Ga enters sphalerite in a more complex way than previously expected. A possible substitution mechanism is $Ga^{3+} + Mn^{2+} + (Cu, Ag)^{+} \leftrightarrow 3Zn^{2+}$. It is thus reasonable to deduce that high Mn contents from the mineralized fluids may facilitate to some degree, the incorporation of Ga into sphalerite. As mentioned above, previous studies suggest that In enters into sphalerite mainly via the coupled substitution mechanism $In^{3+} + (Cu, Ag)^{+} \leftrightarrow 2Zn^{2+}$ (Frenzel et al., 2016). Figure 11D shows that In does not correlate with (Cu, Ag); Figure 11E shows a slightly negative correlation with Fe. Since it is impossible to separate the amounts of (Cu, Ag) involved in In substitution from those involved in other tri- and tetravalent cation substitutions (e.g., Sb^{3+}, Ga^{3+}, As^{3+}, Sn^{4+}, Ge^{4+}), it is not surprising that In and

(Cu, Ag) are not integrated. The reason resulting in a negative correlation between In and Fe is not clear yet and awaits further investigation. Nonetheless, considering the high Fe contents of the Kangjiawan sphalerite, this explains why the Kangjiawan sphalerite is characterized by fewer In contents compared to that from typical high-temperature deposits.

In summary, the incorporation of In, Cd, Ga and Ge into the sphalerite displays different features, mainly depending on the cation oxidation state. Cd enters sphalerite as Cd^{2+} mainly via the isovalent substitution mechanisms ($Cd^{2+} \leftrightarrow Zn^{2+}$). Ga and In are both +3 and coupled substitutions with Cu and Ag can facilitate their incorporation into sphalerite. However, the strong correlation between Ga and Mn indicates that Mn also participated in Ga substitution and a more possible mechanism is $Ga^{3+} + Mn^{2+} + (Cu, Ag)^{+} \leftrightarrow 3Zn^{2+}$. Ge in the Kangjiawan sphalerite is more likely present in the Ge^{4+} state and is likely involved in the following substitution: $Ge^{4+} + \square$ (vacancy) $\leftrightarrow 2Zn^{2+}$ or $Ge^{4+} + 2(Cu, Ag)^{+} \leftrightarrow 3Zn^{2+}$. Cu and Ag maintain charge balance in nearly all substitution mechanisms, with Cu being the primary monovalent cation due to its concentration. The positive correlation between Cu and Ag, as manifested in Figure 11E, likely indicates that their concentrations in fluids were far below the need for charge balance.

Sources of ore-forming materials

The origin of metals and sulfur for sulfides from the Kangjiawan deposit has been extensively discussed. In this study, we report in situ $\delta^{34}S_{V\text{-}CDT}$ values for sphalerite from the main mineralization stage of the Kangjiawan deposit, which range from 1.98 to 5.18 ‰, with an average of 3.63 ± 0.64 ‰ (Fig. 10D). These data are broadly consistent with the previous bulk sulfur isotope data on the Kangjiawan sphalerite from the economic mineralization stage, with the range of $\delta^{34}S_{V\text{-}CDT}$ values from -0.62 to 1.24 ‰ (Li et al., 2021). This sulfur isotope feature is also similar to that of the Kangjiawan pyrite from the same stage, with the range of -1.78 to 2.24 ‰ (Li et al., 2021) and of 1.20 to +3.04 ‰ (Zhang et al., 2024).

In the Kangjiawan deposit, neither sulphate nor oxide was present in the economic mineralization stage. Additionally, as previously noted, Sn, which is sensitive to oxidation, was likely incorporated into sphalerite as Sn^{3+}, suggesting that the oxidation state of the mineralizing fluids was not high. Therefore, the primary sulfur species in the mineralized fluids was most likely H_2S. Li et al. (2021) demonstrated that the mineralized fluids forming the Kangjiawan deposit were weakly acidic. According to Ohmoto. (1979) and recently Belissont et al. (2014), under these physicochemical conditions, sphalerite is characterized by very limited isotopic fractionation, and thus its sulfur isotope data are very close to those of dissolved sulfur (i.e., H_2S) in the mineralized fluids. Combined with data from Li et al. (2021), it can thus be deduced that the $\delta^{34}S_{V\text{-}CDT}$ values of the mineralized fluids range from -0.62 to 5.18 ‰, which is typical of magmatic-derived fluids (-5 to + 5 ‰) (Cooke et al., 2011; Meinert et al., 2005; Shu et al., 2024).

Similar to the element patterns, the Kangjiawan sphalerite does not display systematic variations of sulfur isotopes from core to rim at the micrometer scale. A correlation between sulfur isotope values and types of colour banding is also not observed, even though sulfur isotopes slightly fluctuated with colour banding. This indicates that the fluids (and probably also metals) from the economic mineralization stage are derived from a consistent magmatic source without an evident involvement from other reservoirs. The slight variations of sulfur isotopes could be explained by the competitive growth of other sulfides (e.g., pyrite and galena) at or near the growing sphalerite crystal edges, due to isotope fractionations between co-existing sulfur-bearing compounds (Rye and Ohmoto, 1974). We suggest that the stable supply of fluids and metals in the economic mineralization stage accounts for the formation of a large Cd-, Ga- and In-bearing ore-forming system in the Kangjiawan deposit. Nonetheless, as reported in many magmatic-hydrothermal systems, the sources of metal and fluids may change from the early to the late stages, with the metals and fluids from the late stage illustrating more influence from wall-rocks and meteoric fluids (Seal, 2006; Shu et al., 2021). This is also the case of the Kangjiawan deposit, where Li et al. (2021) showed that sphalerite from the late stage is characterized by $\delta^{34}S_{V\text{-}CDT}$ values of -6.88 to -5.79 ‰, much lower than those from the economic mineralization stage. However, whether this signature is the result of the involvement of wall-rocks or meteoric fluids is not clear yet and awaits further investigation.

Comparison to sphalerites from the MVT and skarn deposits

To illustrate the enrichment degree of Ga, In and Cd, as well as some other related elements (e.g., Ge, Cu, In) in the Kangjiawan sphalerite, we compared our sphalerite trace and minor element data with those from the MVT and skarn deposits (Table S4), compiled by Frenzel et al. (2016) and more recently by Meng et al. (2024) and Tan et al. (2024). MVT and high-temperature hydrothermal replacement (HTHR) sphalerites represent those forming from low- and high-temperature end-members, respectively, whereas temperature has been proposed as one of the most important factors in controlling enrichment of high-tech critical metals (e.g., Ge and Ga) (Belissont et al., 2016; Frenzel et al., 2016).

The Kangjiawan sphalerite shares many features of global skarn deposits, including lower Ge and Ag contents, and higher Fe, In, Cd and Mn contents compared to MVT deposits (Fig. 12). This is consistent with previous studies which illustrate that the main primary sources of Ge are sphalerite derived from low-temperature hydrothermal systems (Belissont et al., 2016; Frenzel et al., 2016), and that sphalerite can be used to discriminate among ore genetic types. For example, using machine learning methods, Tan et al. (2024) showed that high Ge and low Mn contents of sphalerite can act as robust indicators to distinguish the MVT from skarn deposits. These features are also consistent with the study by Frenzel et al. (2016), which concluded that concentrations of Fe, Mn and In increase, while Ge contents decrease with increasing formation temperature. However, different from the generally low Ga contents from global skarn deposits (1.8-5.3 ppm), the Kangjiawan sphalerite is characterized by higher Ga contents (up to 41.4 ppm), similar to that from MVT deposits. Cd (and also Cu) concentrations of the Kangjiawan sphalerite (2448.7-4586.5 ppm, 20.6-6375.0 ppm) are almost consistent with those from global skarn deposits (2400-3600 ppm; 37-530 ppm) and MVT deposits (2700-4800 ppm; 210-560 ppm). The Kanjiawan sphalerite is characterized by much higher Sn contents (up to 2926 ppm), whereas the global compilation does not show noticeable distinctions of Sn contents from skarn and MVT deposits.

The above results demonstrate that although generally sphalerites from skarn deposits are characterized by lower Ga and Ge contents than those from the MVT deposits, skarn deposits, especially those from South China, could still be promising exploration targets for Ga (Burke and Kieft, 1980; Frenzel et al., 2014; 2016; Cook et al., 2011). Kangjiawan is such a deposit with its sphalerite characterized by enriched Ga contents, which is similar to MVT deposits. Besides, skarn and MVT deposits have the same potential to form economic Cd mineralization (⩾0.2% Cd in lead-zinc ores according to the Mineral Resources Industry Requirements Manual, 2022), whereas skarn deposits are more important for In exploration. The enrichment of the high-tech critical metals should be highly related to the temperature of ore-forming fluids, as emphasized in many studies. The relative enrichment of Ga and depletion of In of the Kangjiawan deposit compared to common skarn deposits is not clear. But as previously discussed, the in situ sulfur isotope analysis of the Kangjiawan deposit suggests that the sulfur of the ore-forming fluids originated from a single magmatic source without any observable disturbance of external fluids. Considering that the significantly lower initial In contents of the lowly evolved magma than the highly evolved one and there is no additional contribution of Ga from wall-rocks (Gion et al., 2019; Wang et al., 2021; Zhang, 2019), it is reasonable to deduce that the high Ga/In ratios of sphalerite in the Kangjiawan magmatic-hydrothermal systems could probably be inherited from the parental magmas (e.g., the Shuikoushan granodiorite characterized by Ga contents up to 25.6 ppm and In contents of 0.02-0.43 ppm; Zhao, 2017; Qin et al., 2023).

CONCLUSIONS

Based on combined analyses of major and trace elements, sulfur isotopes, and elemental mapping on banded sphalerite from the Kangjiawan skarn Pb-Zn deposit in South China, three typical optical zoning bands are identified. The darker bands display higher Fe, Mn and Cu concentrations than the light bands. Spot concentrations for Cd, Ga and In are high, which, however, do not correlate with the colour banding. Moreover, minor and trace elements do not display any systematic increase/decrease from core to rim at the micrometer-scale, which indicates that microscale processes at or near the growing crystal edges account for the observed compositional zoning. In situ sulfur isotope analyses for the Kangjiawan sphalerite suggest a consistent fluid source, typical of magmatic-derived fluids. Moreover, sulfur isotopes do not display systematic variations from core to rim, further highlighting intrinsic, self-organized processes. The elemental correlation analyses suggest the isovalent substitution $Cd^{2+} \leftrightarrow Zn^{2+}$, and coupled substitution $In^{3+} + (Cu, Ag)^{+} \leftrightarrow 2Zn^{2+}$. A strong correlation between Ga and Mn is manifested, suggesting a more complex mechanism $Ga^{3+} + Mn^{2+} + (Cu, Ag)^{+} \leftrightarrow 3Zn^{2+}$. In comparison with sphalerites from global skarn and MVT deposits, this study advocates that low-temperature MVT deposits are more promising for Ga and Ge exploration and high-temperature skarn deposits are more important for In exploration, although there is no distinction in forming economic Cd mineralization. Strikingly, the Kangjiawan sphalerites are characterized by high Ga contents, similar to those from MVT deposits. These findings highlight that skarn Pb-Zn deposits from South China are promising for not only Cd and In, but also Ga exploration.

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FIGURE CAPTIONS

Figure 1 A. Simplified geological map of China. B. Simplified distribution map of W, Sn, Cu, Pb, Zn, Au deposits of South China with the location of the Qin-Hang belt ore field shown (modified after Yuan et al., 2018).

Figure 2 A. Geological map of the Shuikoushan ore field. B. Geological map of Nos. 9-12 faces of the Kangjiawan skarn Pb-Zn deposit (modified after Gong, 2012). C. Stratigraphic column diagram of the Shuikoushan ore field (modified after Qin et al., 2022).

Figure 3 Field photographs (A-F) and micrographs (G-M) of typical hydrothermal zones and Pb-Zn ores from the Kangjiawan deposit. A & E. Garnet skarn. B. Silicified breccia belts were cut cross by quartz-pyrite veins. C. Lead-zinc ore bodies were altered by calcite veins. D & F. Ores from the drill core CK103-1. G. Garnet skarn was replaced by chlorite (plane-polarized light). H. Disseminated pyrite and chalcopyrite in garnet skarn (reflected light). I. Silicified breccia was cemented by veined pyrite (reflected light). J. Paragenetic sphalerite, pyrite and galena in lead-zinc ore bodies (reflected light). K-L. Co-existing sphalerite and arsenopyrite in a calcite vein (plane-polarized light and reflected light). M. Typical banded sphalerite in a calcite vein (plane-polarized light). Abbreviations: Grt, Garnet; Mgt, Magnetite; As, Arsenopyrite; Sp, sphalerite; Gn, Galena; Py, Pyrite; Cpy, Chalcopyrite; Qtz, Quartz; Cc, Calcite; Chl, Chlorite.

Figure 4 Minerals sequence of Kangjiawan Pb-Zn polymetallogenic deposit.

Figure 5 Micrographs of banded sphalerite in the Kangjiawan deposit (plane-polarized light). A. Clusters of banded sphalerite with four grains analyzed in this study. B-I. Amplified sphalerite grains of Sph-1 to Sph-4 indicated in A. R1 to R2 refer to band types (see text for more details).

Figure 6 LA-ICP-MS elemental mapping results for the Kangjiawan sphalerite (Sph-2). A. Studied grain Sph-2 under plane-polarized light. B-L. Mapping results for Mn, Cu, Zn, Fe, Pb, Ga, Ge, In, Cd, Ag and Sb, respectively. Note that the darker the color, the higher the element content. The white boxes are provided for quick location among different figures.

Figure 7 Sphalerite grains used for EMPA analyses (A-C), as well as the corresponding elemental variations.

Figure 8 Histograms of trace and minor element compositions of banded sphalerite from the Kangjiawan deposit. A. In. B. Ga. C. Ge. D. Sb. E. Sn. F. Ag. G. Au. H. Mn. I. Cu. J. Pb. K. As.

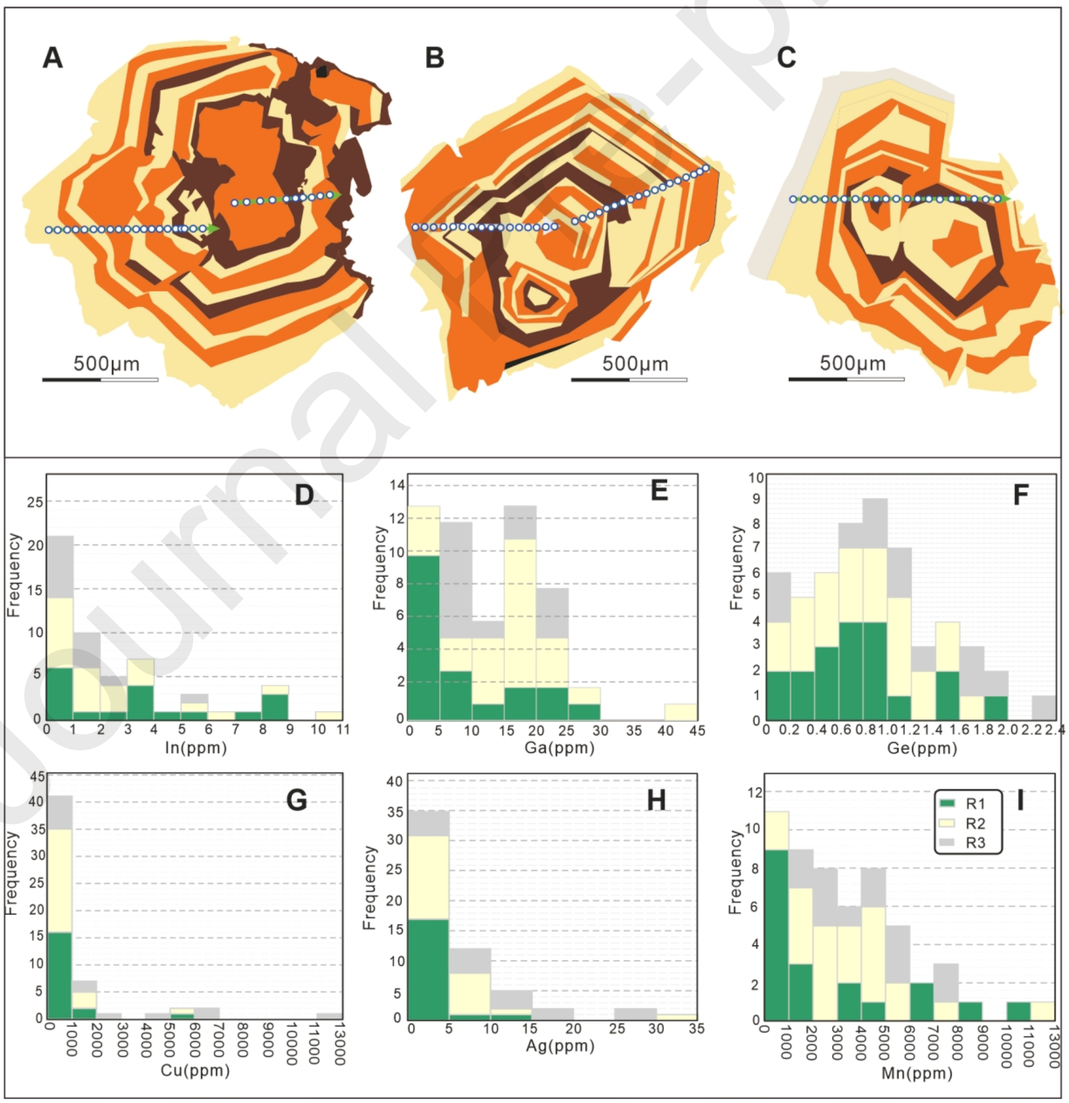
Figure 9 PCA cluster diagrams for sphalerite from Kangjiawan deposit.

Figure 10 In situ sulfur isotope results for the Kangjiawan sphalerite (Sph-2). A-C. $\delta^{34}S_{V\text{-}CDT}$ values for bands R1-R3, respectively. D. Variation of the averaged $\delta^{34}S_{V\text{-}CDT}$ values for each band.

Figure 11 Binary correlation plots for the Kangjiawan sphalerite. A. Cu + Ag versus Ga + In + Ge + Sn +Sb +As. B. Cu + Ag versus Sn. C. Ga versus Mn. D. Cu + Ag versus In. E. Fe/10000 versus In. F. Cu versus Ag. The units for x- and y-axis are μmol/g. Diagonal lines represent 1:1 correlations between concentrations (μmol/g).

Figure 12 Boxplots of concentrations of nine common elements of sphalerite in different deposit types. A. Ga. B. Ge. C. In. D. Cd. E. Cu. F. Fe. G. Ag. H. Mn. I. Sn. Skarn and MVT sphalerite data are derived from global compilation (see Table S4 in Supplemental Material).

Graphical Abstract



Highlights

1.There are 3 typical optical zoning bands can be identified in Kangjiawan deposit with the homogeneously distributed of Cd and distribution patterns of Ga and In.

2. The banded sphalerite should be formed by intrinsic, self-organized processes.

3. The skarn Pb-Zn deposits of South China are for not only Cd (and probably In) exploration, but also Ga exploration.

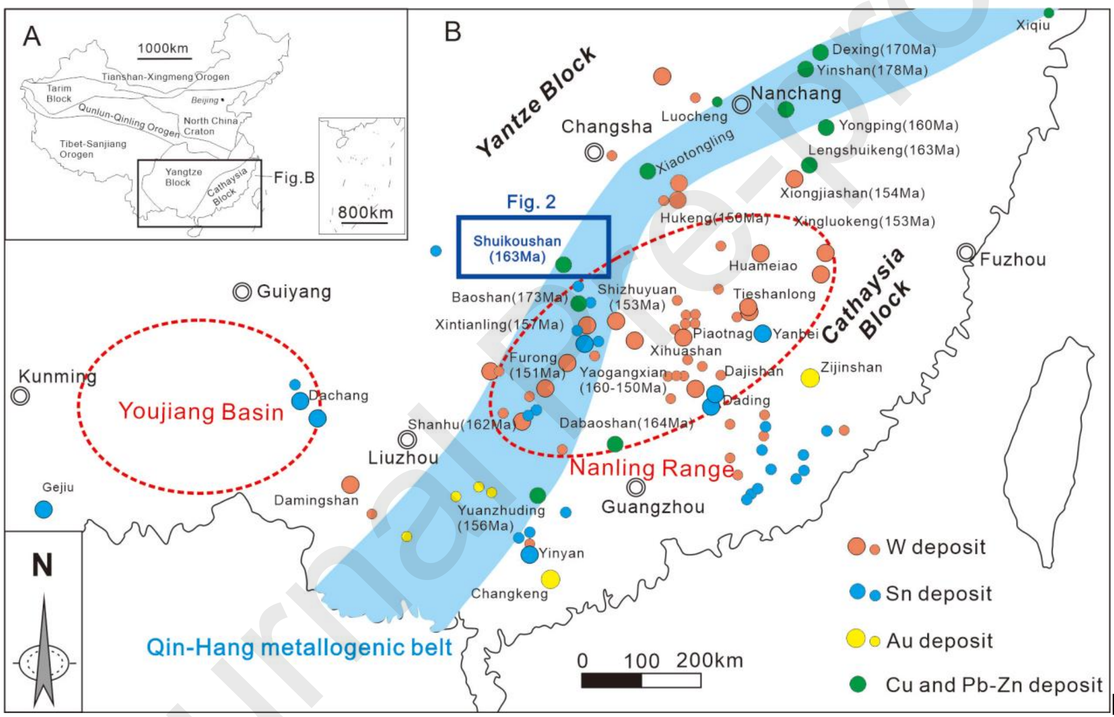


Figure 1

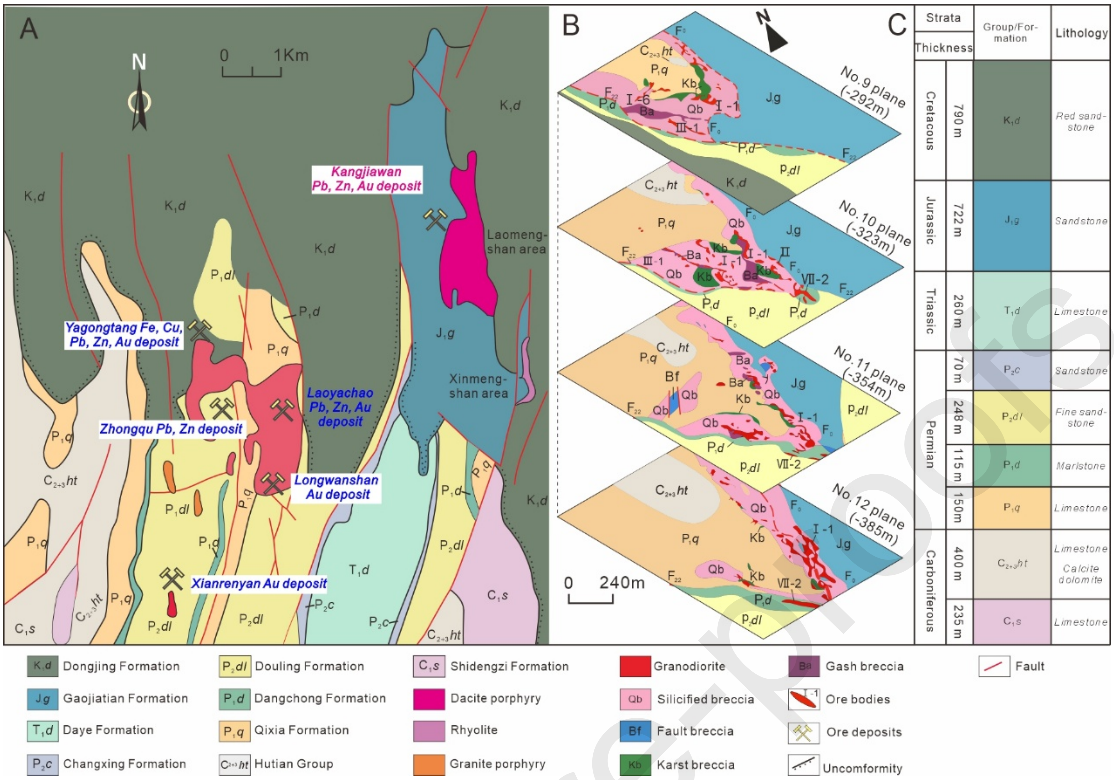

A
N
0 1Km
Kangjiawan Pb, Zn, Au deposit
Laomeng-shan area
Yagongtang Fe, Cu, Pb, Zn, Au deposit
Zhongqu Pb, Zn deposit
Laoyachao Pb, Zn, Au deposit
Longwanshan Au deposit
Xianrenyan Au deposit
Xinmeng-shan area
B
C
No.9 plane (-292m)
No.10 plane (-323m)
No.11 plane (-354m)
No.12 plane (-385m)
0 240m
Strata
Thickness
Group/Formation
Lithology
Cretacous
Jurassic
Triassic
Permian
Carboniferous
790 m
722 m
260 m
70 m
248 m
115 m
150m
400 m
235 m
Red sand-stone
Sandstone
Limestone
Sandstone
Fine sand-stone
Marlstone
Limestone
Limestone
Calcite dolomite
Limestone
Dongjing Formation
Gaojiatian Formation
Daye Formation
Changxing Formation
Douling Formation
Dangchong Formation
Qixia Formation
Hutian Group
Shidengzi Formation
Dacite porphyry
Rhyolite
Granite porphyry
Granodiorite
Silicified breccia
Fault breccia
Karst breccia
Gash breccia
Ore bodies
Ore deposits
Uncomformity
Fault

Figure 2

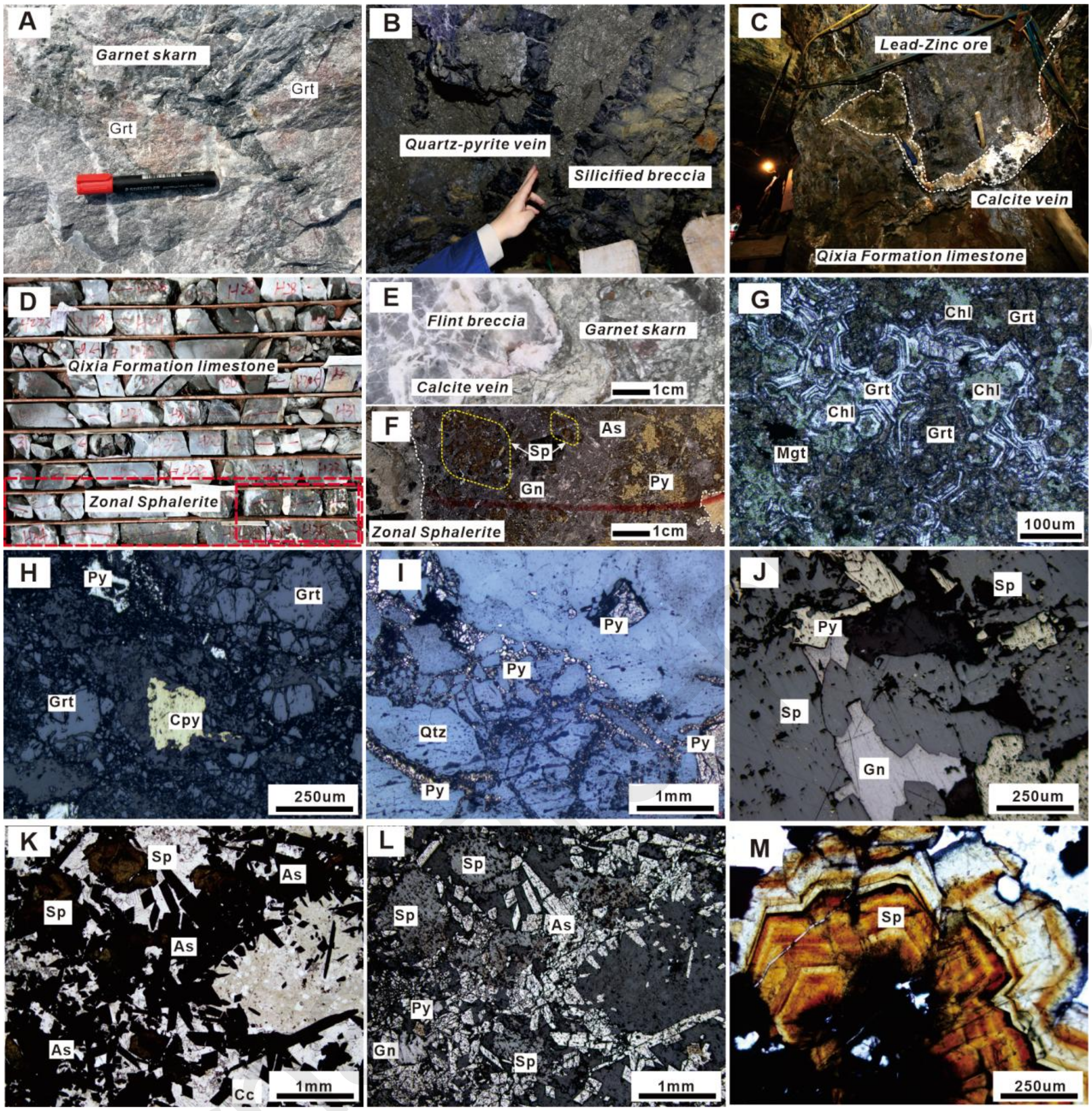
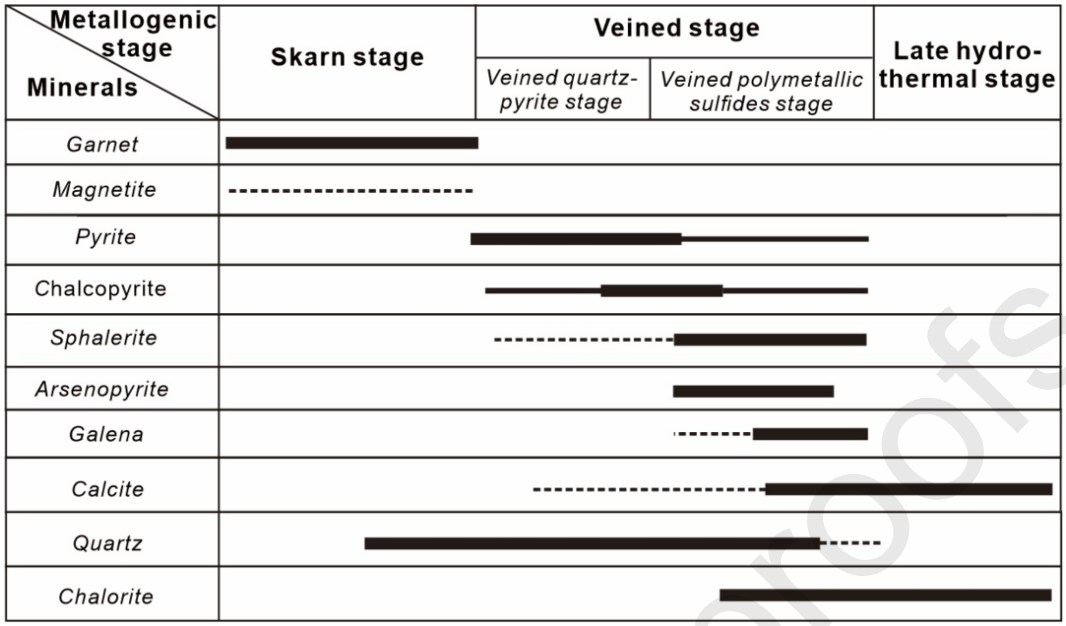

A
Garnet skarn
Grt
Grt
B
Quartz-pyrite vein
Silicified breccia
C
Lead-Zinc ore
Calcite vein
Qixia Formation limestone
D
Qixia Formation limestone
Zonal Sphalerite
E
Flint breccia
Garnet skarn
Calcite vein
1cm
F
Sp
As
Gn
Py
Zonal Sphalerite
1cm
G
Chl
Grt
Grt
Chl
Chl
Grt
Mgt
100um
H
Py
Grt
Grt
Cpy
250um
I
Py
Py
Qtz
Py
Py
1mm
J
Sp
Py
Sp
Gn
250um
K
Sp
As
Sp
As
As
Cc
1mm
L
Sp
Sp
As
Py
Gn
Sp
1mm
M
Sp
250um

Figure 3



Metallogenic stage
Minerals
Skarn stage
Veined stage
Veined quartz-pyrite stage
Veined polymetallic sulfides stage
Late hydro-thermal stage
Garnet
Magnetite
Pyrite
Chalcopyrite
Sphalerite
Arsenopyrite
Galena
Calcite
Quartz
Chalorite

Figure 4

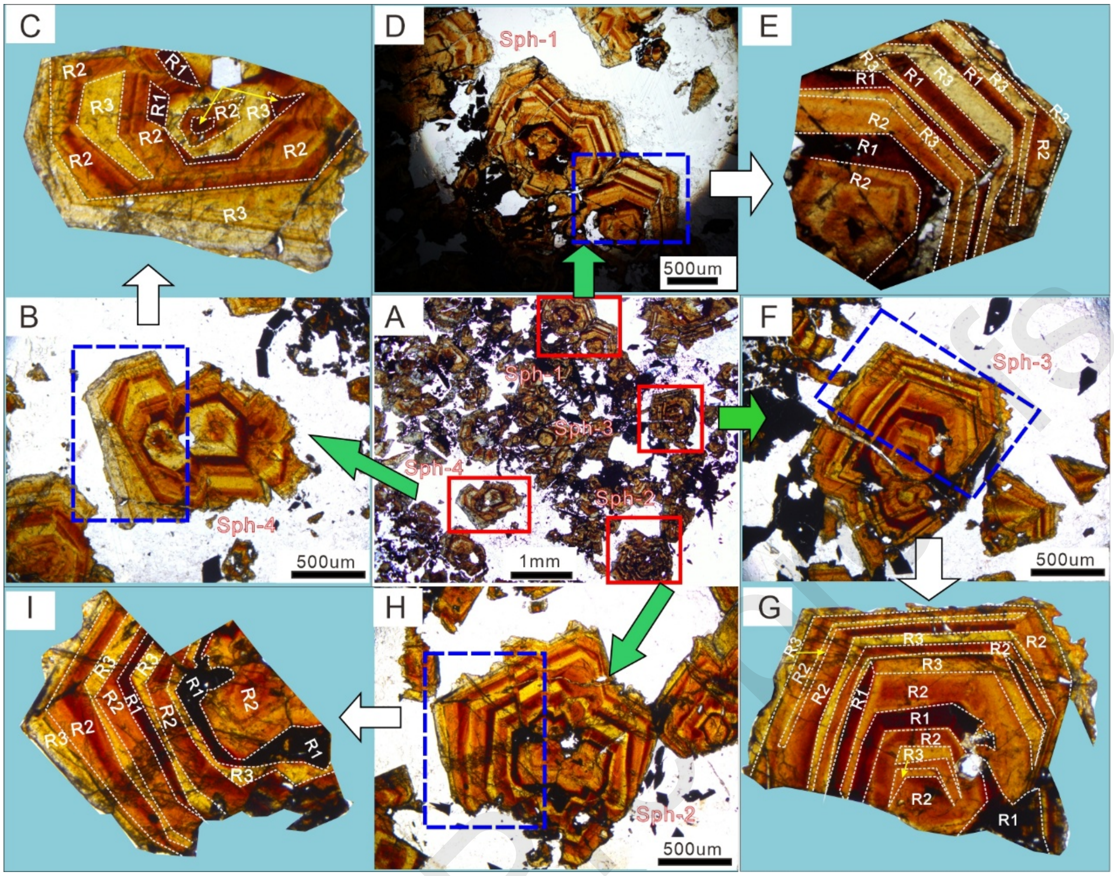


Figure 5

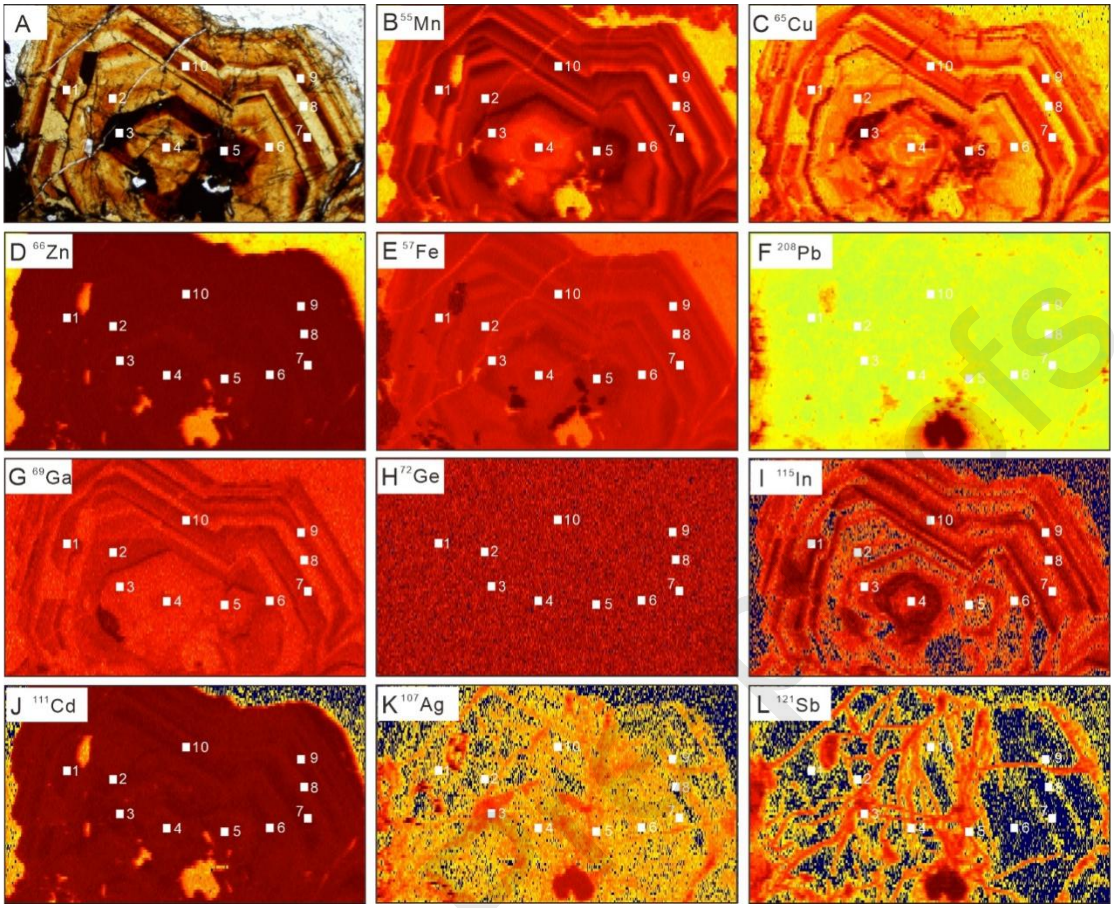

A
B ^{55}Mn
C ^{65}Cu
D ^{66}Zn
E ^{57}Fe
F ^{208}Pb
G ^{69}Ga
H ^{72}Ge
I ^{115}In
J ^{111}Cd
K ^{107}Ag
L ^{121}Sb
1
2
3
4
5
6
7
8
9
10

Figure 6

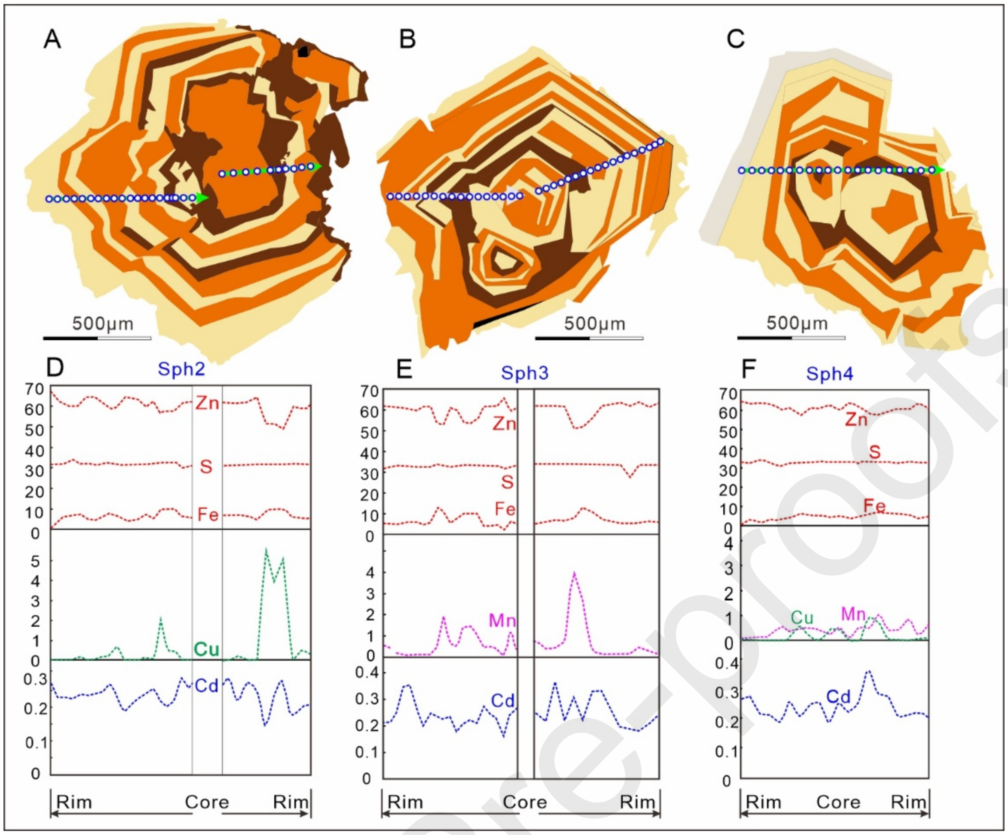
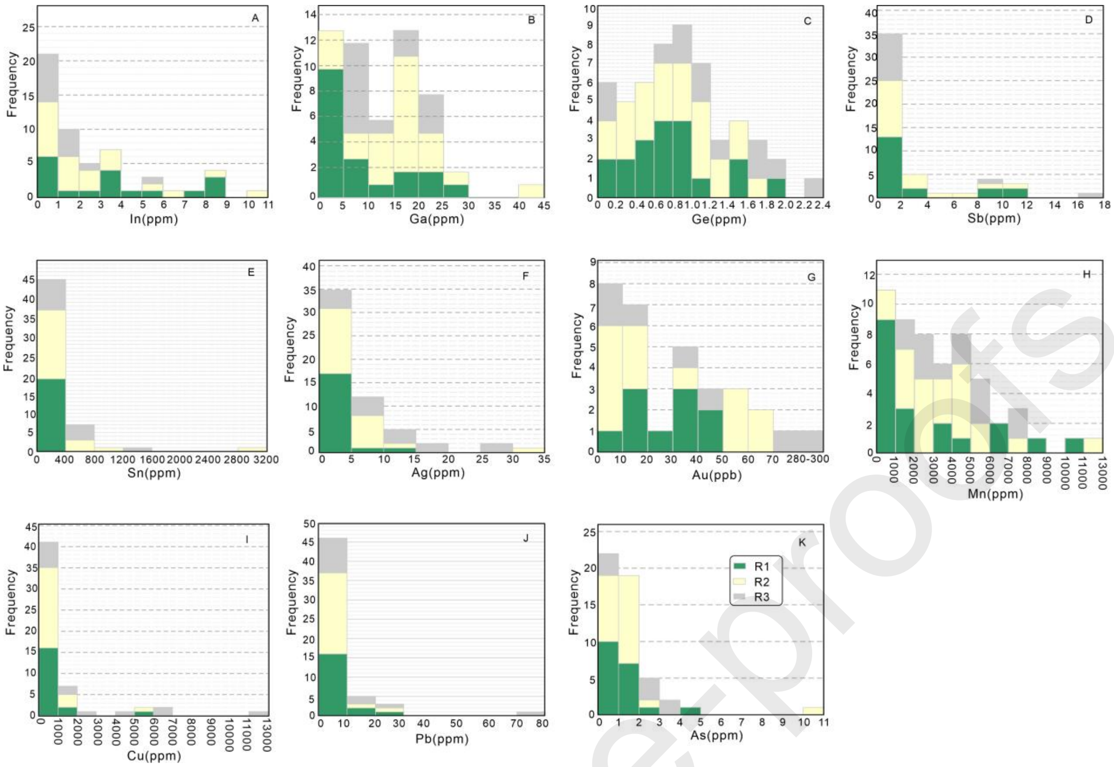

A
B
C
500μm
500μm
500μm
D
Sph2
E
Sph3
F
Sph4
Zn
S
Fe
Cu
Mn
Cd
Rim
Core
Rim

Figure 7



A
B
C
D
E
F
G
H
I
J
K
Frequency
In(ppm)
Ga(ppm)
Ge(ppm)
Sb(ppm)
Sn(ppm)
Ag(ppm)
Au(ppb)
Mn(ppm)
Cu(ppm)
Pb(ppm)
As(ppm)
R1
R2
R3

Figure 8

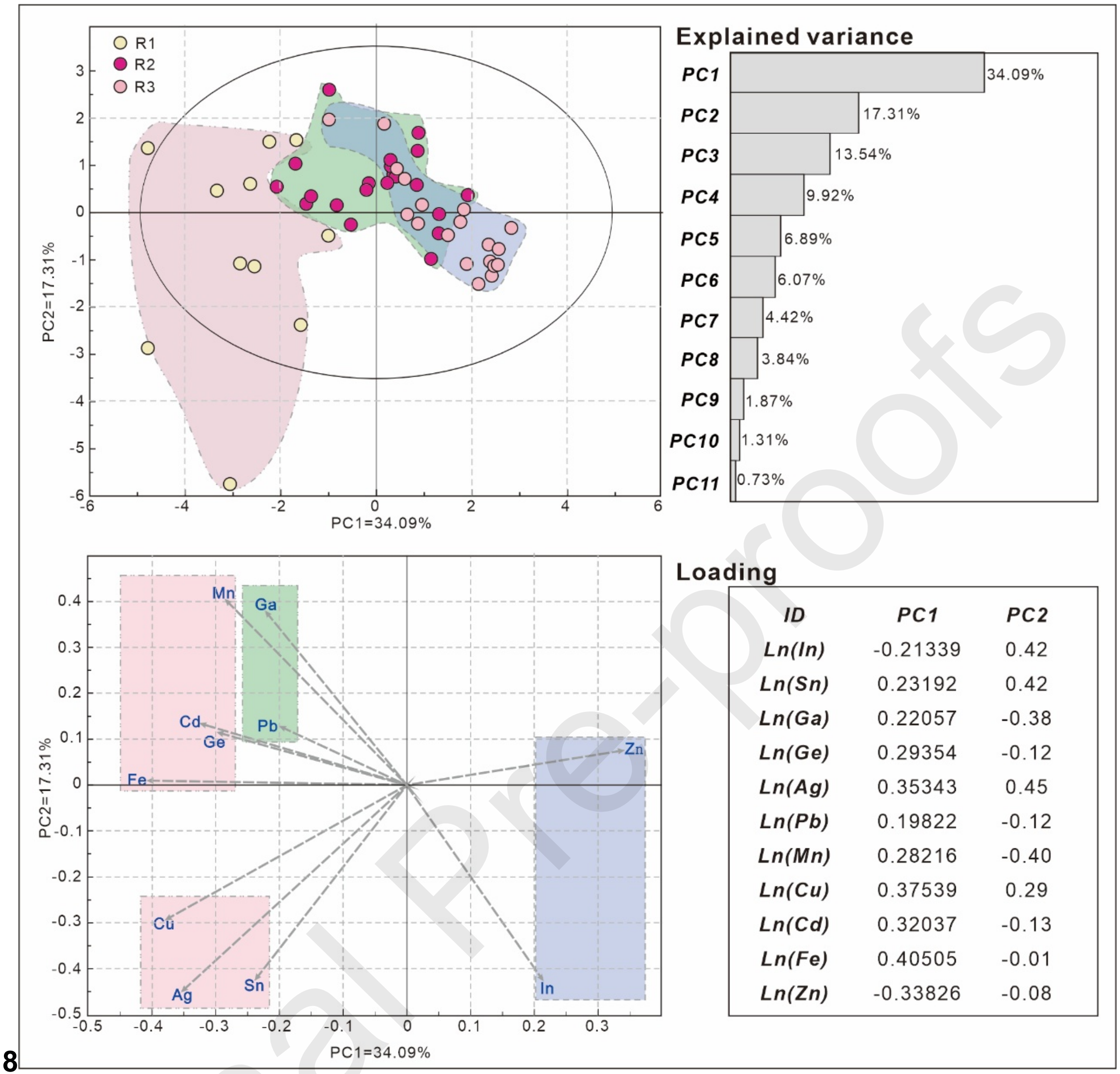


Figure 9

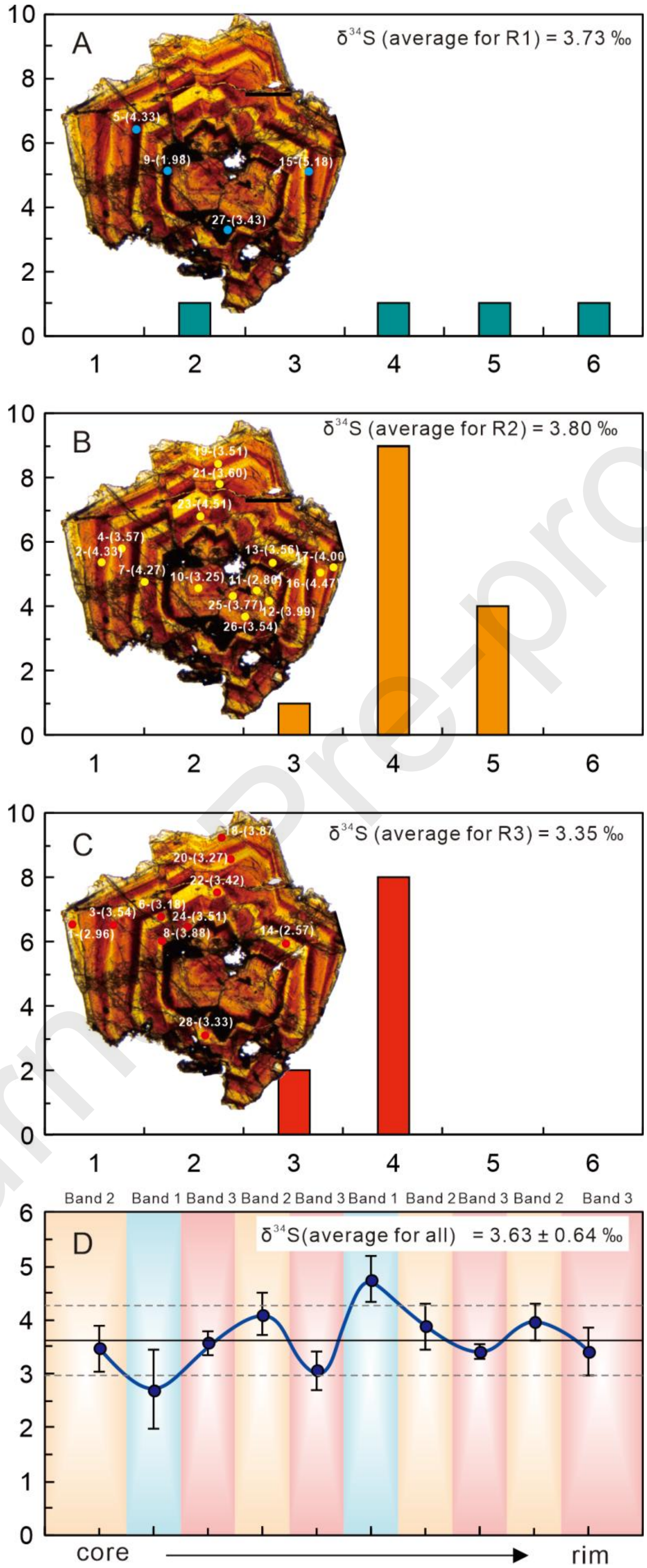

A
δ34S (average for R1) = 3.73 ‰
5-(4.33)
9-(1.98)
15-(5.18)
27-(3.43)
B
δ34S (average for R2) = 3.80 ‰
19-(3.51)
21-(3.60)
23-(4.51)
4-(3.57)
2-(4.33)
7-(4.27)
10-(3.25)
11-(2.80)
13-(3.56)
17-(4.00)
16-(4.47)
25-(3.77)
12-(3.99)
26-(3.54)
C
δ34S (average for R3) = 3.35 ‰
18-(3.87)
20-(3.27)
22-(3.42)
3-(3.54)
6-(3.18)
24-(3.51)
1-(2.96)
8-(3.88)
14-(2.57)
28-(3.33)
Band 2
Band 1
Band 3
Band 2
Band 3
Band 1
Band 2
Band 3
Band 2
Band 3
D
δ34S(average for all) = 3.63 ± 0.64 ‰
core
rim

Figure 10

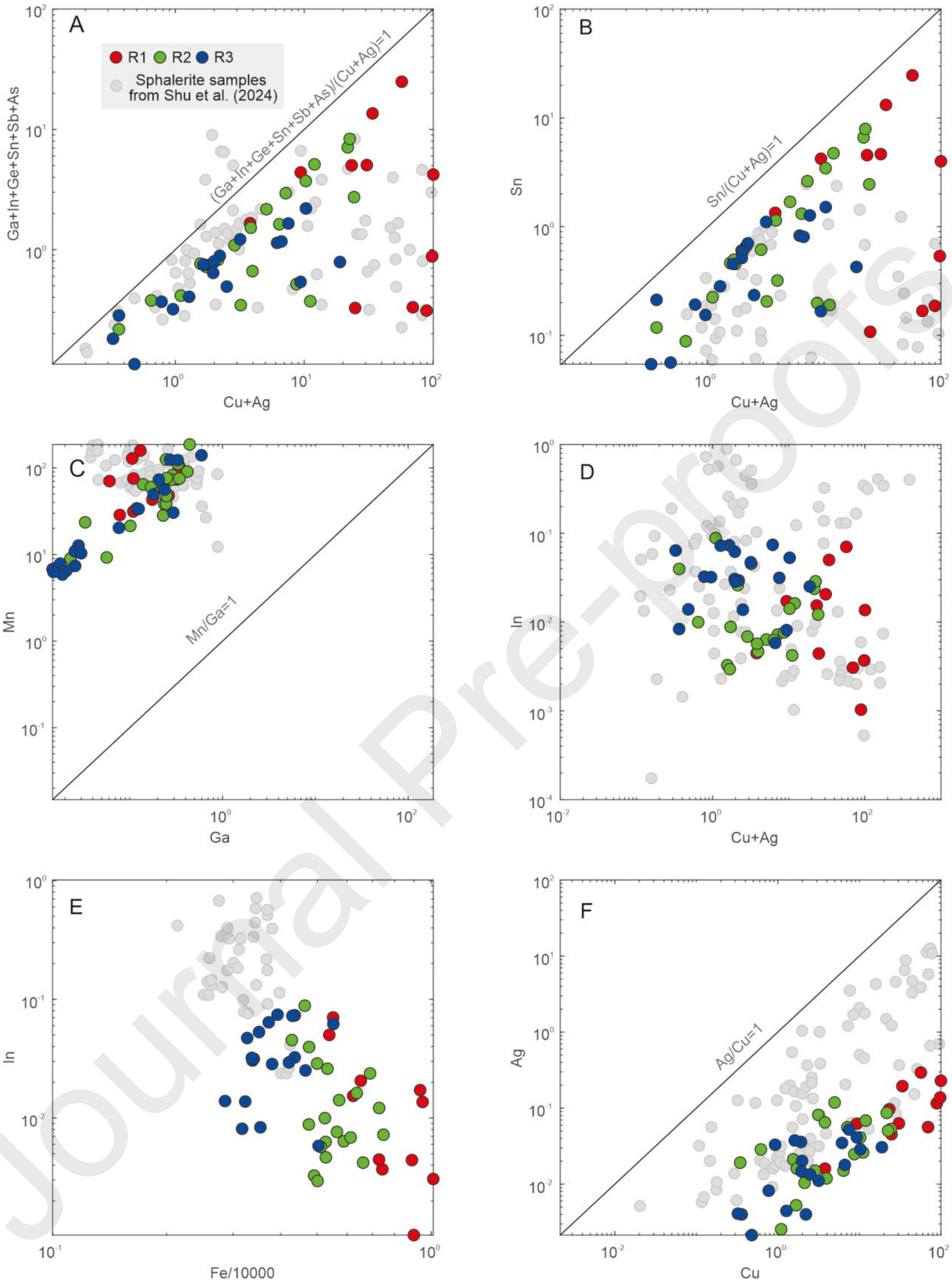

A
R1 R2 R3
Sphalerite samples from Shu et al. (2024)
(Ga+In+Ge+Sn+Sb+As)/(Cu+Ag)=1
Ga+In+Ge+Sn+Sb+As
Cu+Ag
B
Sn/(Cu+Ag)=1
Sn
Cu+Ag
C
Mn/Ga=1
Mn
Ga
D
In
Cu+Ag
E
In
Fe/10000
F
Ag/Cu=1
Ag
Cu

Figure 11

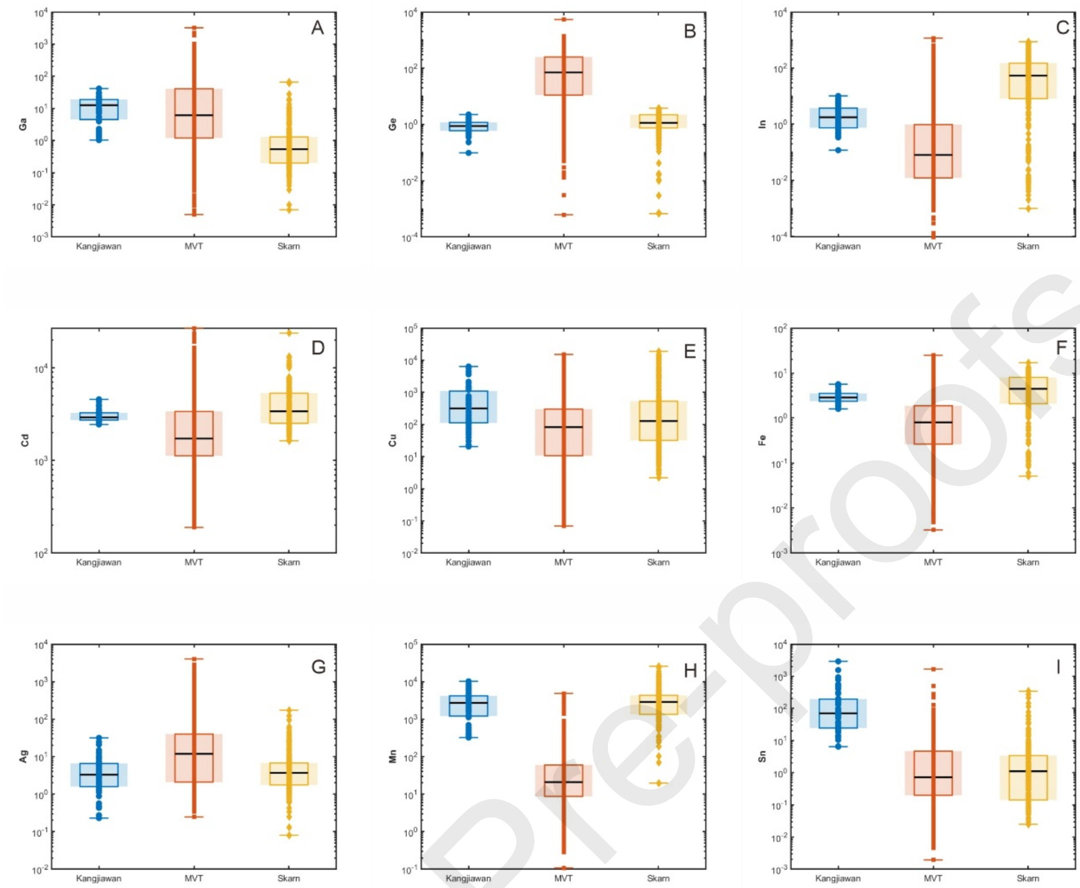


Figure 12